Quantitative Comparisons

ANSWER KEY

Numeric comparisons

1 Column A: 7×8. Column B: $6 \times 9 + 2$. Compare the two columns: which is greater, are they equal, or is it impossible to determine?

$A = 56$. $B = 54 + 2 = 56$. $A = B$.

2 Column A: $123 + 456$. Column B: $700 - 120$. Compare the two columns.

$A = 579$. $B = 580$. $B > A$.

3 Column A: 9^2. Column B: $8^2 + 15$. Compare the two columns.

$A = 81$. $B = 64 + 15 = 79$. $A > B$.

4 Column A: $(-3)^2$. Column B: -3^2. Compare the two columns.

$A = (-3) \times (-3) = 9$. $B = -(3^2) = -9$ (no parentheses around the base). $A > B$.

Fraction and decimal comparisons

5 Column A: $\frac{3}{7}$. Column B: 0.43. Compare the two columns.

$\frac{3}{7} \approx 0.4286$, which is less than 0.43. $B > A$.

6 Column A: $\frac{7}{12}$. Column B: 0.58. Compare the two columns.

$\frac{7}{12} \approx 0.5833$, which is greater than 0.58. $A > B$.

7 Column A: $2\frac{3}{8}$. Column B: $\frac{19}{8}$. Compare the two columns.

$2\frac{3}{8} = \frac{19}{8}$. $A = B$.

8 Column A: 0.6. Column B: $\frac{5}{9}$. Compare the two columns.

$\frac{5}{9} \approx 0.5556$, which is less than 0.6. $A > B$.

Percent comparisons

9 Column A: 25% of 160. Column B: 20% of 200. Compare the two columns.

$A = 0.25 \times 160 = 40$. $B = 0.20 \times 200 = 40$. $A = B$.

10 Column A: 30% of 90. Column B: 40% of 60. Compare the two columns.

$A = 0.3 \times 90 = 27$. $B = 0.4 \times 60 = 24$. $A > B$.

11 Column A: the result of increasing 100 by 10% twice in a row. Column B: the result of increasing 100 by 20% once. Compare the two columns.

$A = 100 \times 1.1 \times 1.1 = 121$. $B = 100 \times 1.2 = 120$. $A > B$.

12 Column A: 15% of 80. Column B: 12% of 100. Compare the two columns.
$A = 0.15 \times 80 = 12$. $B = 0.12 \times 100 = 12$. $A = B$.

Exponent and root comparisons

13 Column A: $\sqrt{50}$. Column B: 7.1. Compare the two columns.
$\sqrt{50} \approx 7.07$, which is less than 7.1. $B > A$.

14 Column A: 2^{10}. Column B: 10^3. Compare the two columns.
$A = 1{,}024$. $B = 1{,}000$. $A > B$.

15 Column A: $\sqrt{0.09}$. Column B: 0.3. Compare the two columns.
$\sqrt{0.09} = 0.3$. $A = B$.

16 Column A: 3^4. Column B: 4^3. Compare the two columns.
$A = 81$. $B = 64$. $A > B$.

Algebraic comparisons at a fixed value

17 Given $x = 5$. Column A: $2x + 3$. Column B: $x + 8$. Compare the two columns.
$A = 2(5) + 3 = 13$. $B = 5 + 8 = 13$. $A = B$.

18 Given $x = 4$. Column A: $(x + 2)^2$. Column B: $x^2 + 4x + 4$. Compare the two columns.
$(x + 2)^2$ expands to $x^2 + 4x + 4$ for every value of x, so the columns are always equal. $A = B$.

19 Given $n = -2$. Column A: n^2. Column B: n^3. Compare the two columns.
$A = (-2)^2 = 4$. $B = (-2)^3 = -8$. $A > B$.

20 Given $x = 3$. Column A: $5x - 2$. Column B: $2(x + 5)$. Compare the two columns.
$A = 5(3) - 2 = 13$. $B = 2(3 + 5) = 16$. $B > A$.

Comparisons requiring testing values

21 For any real number x: Column A: x. Column B: x^2. Compare the two columns.
If $x = 2$: $B > A$. If $x = \frac{1}{2}$: $A > B$. If $x = 1$: $A = B$. The relationship changes, so it cannot be determined.

22 For any real number x: Column A: $x + 5$. Column B: $2x$. Compare the two columns.
If $x = 5$: $A = B$. If $x = 0$: $A > B$. If $x = 10$: $B > A$. The relationship changes, so it cannot be determined.

23 For any real number x: Column A: $|x|$. Column B: x. Compare the two columns.

If $x \geq 0$: $A = B$. If $x < 0$ (e.g. $x = -3$): $A = 3$, $B = -3$, so $A > B$. Since the relationship is not the same in every case, it cannot be determined.

24 Given only that n is an integer: Column A: n^2. Column B: 4. Compare the two columns.

If $n = 1$: $A = 1 < 4$. If $n = 3$: $A = 9 > 4$. The relationship changes, so it cannot be determined.

25 For any real number x: Column A: $(x - 3)^2$. Column B: $x^2 - 6x + 9$. Compare the two columns.

Expanding: $(x - 3)^2 = x^2 - 6x + 9$ for every value of x, so the columns are always equal. $A = B$.

Geometry comparisons

26 Column A: the perimeter of a square with side 6 cm. Column B: the perimeter of a rectangle with length 8 cm and width 3 cm. Compare the two columns.

$A = 4 \times 6 = 24$ cm. $B = 2(8 + 3) = 22$ cm. $A > B$.

27 Column A: the area of a circle with radius 4 cm (use $\pi \approx 3.14$). Column B: the area of a square with side 7 cm. Compare the two columns.

$A = 3.14 \times 4^2 = 3.14 \times 16 = 50.24$ cm^2. $B = 7^2 = 49$ cm^2. $A > B$.

28 Column A: the sum of the interior angles of a pentagon. Column B: the sum of the interior angles of two triangles. Compare the two columns.

$A = (5 - 2) \times 180^\circ = 540^\circ$. $B = 2 \times 180^\circ = 360^\circ$. $A > B$.

29 Column A: the hypotenuse of a right triangle with legs 5 and 12. Column B: 14. Compare the two columns.

$A = \sqrt{5^2 + 12^2} = \sqrt{169} = 13$. $B = 14$. $B > A$.

Ratio and proportion comparisons

30 Column A: the value of x, if $x : 12 = 3 : 4$. Column B: 10. Compare the two columns.

$x = \frac{3 \times 12}{4} = 9$. $B = 10$. $B > A$.

31 A recipe uses flour and sugar in the ratio $3 : 1$. Column A: the sugar needed for 240 g of flour. Column B: 90 g. Compare the two columns.

Sugar $= 240 \div 3 = 80$ g. $B = 90$ g. $B > A$.

32 Column A: the sum of the parts in the simplified form of the ratio $18 : 24$. Column B: 8. Compare the two columns.

$18 : 24$ simplifies to $3 : 4$, so the sum of parts is $3 + 4 = 7$. $B = 8$. $B > A$.

33 Two numbers are in the ratio $5:3$, and their sum is 96. Column A: the larger number. Column B: 61. Compare the two columns.

Total parts $= 5 + 3 = 8$, each part $= 12$. Larger number $= 5 \times 12 = 60$. $B = 61$. $B > A$.

Data and probability comparisons

34 A data set is $4, 7, 7, 9, 13$. Column A: the mean. Column B: the median. Compare the two columns.

Mean $= \frac{4+7+7+9+13}{5} = \frac{40}{5} = 8$. Median (middle value) $= 7$. $A > B$.

35 Column A: the range of the data set $3, 8, 15, 22$. Column B: 20. Compare the two columns.

Range $= 22 - 3 = 19$. $B = 20$. $B > A$.

36 A bag contains 3 red balls and 5 blue balls. Column A: the probability of drawing a red ball. Column B: the probability of drawing a blue ball. Compare the two columns.

$P(\text{red}) = \frac{3}{8}$. $P(\text{blue}) = \frac{5}{8}$. $B > A$.

Word-problem comparisons

37 Column A: the total cost of 4 items at 15 riyals each, after a 10% discount on the total. Column B: 55 riyals. Compare the two columns.

Total before discount $= 4 \times 15 = 60$ riyals. After discount: $60 \times 0.9 = 54$ riyals. $B = 55$. $B > A$.

38 Ali's age is 3 years more than twice Sara's age. Sara is 8. Column A: Ali's age. Column B: 20. Compare the two columns.

Ali $= 2(8) + 3 = 19$. $B = 20$. $B > A$.

39 Column A: the simple interest earned on 2,000 riyals at 5% per year for 3 years. Column B: 350 riyals. Compare the two columns.

$I = 2{,}000 \times 0.05 \times 3 = 300$ riyals. $B = 350$. $B > A$.

40 A car travels 240 km using 20 liters of fuel. Column A: the distance travelled per liter. Column B: 12 km per liter. Compare the two columns.

Distance per liter $= 240 \div 20 = 12$ km/liter. $B = 12$. $A = B$.

Comparisons with inequalities and ranges

41 Given that $5 < x < 9$. Column A: x. Column B: 7. Compare the two columns.

If $x = 6$: $A < B$. If $x = 8$: $A > B$. The relationship changes, so it cannot be determined.

42 Given that $x > 10$. Column A: x. Column B: 10. Compare the two columns.

Since x is always greater than 10, $A > B$ in every case.

43 Given that $0 < x < 1$. Column A: x. Column B: x^2. Compare the two columns.

For any value strictly between 0 and 1 (e.g. $x = 0.5$: $x^2 = 0.25 < 0.5$), squaring makes the value smaller. $A > B$ in every case.

44 Given that $-5 \leq x \leq -1$. Column A: x^2. Column B: 10. Compare the two columns.

If $x = -1$: $A = 1 < 10$. If $x = -5$: $A = 25 > 10$. The relationship changes, so it cannot be determined.

More geometry comparisons

45 Column A: the volume of a cube with edge 4 cm. Column B: the volume of a rectangular box $3 \text{ cm} \times 4 \text{ cm} \times 6 \text{ cm}$. Compare the two columns.

$A = 4^3 = 64 \text{ cm}^3$. $B = 3 \times 4 \times 6 = 72 \text{ cm}^3$. $B > A$.

46 Column A: the circumference of a circle with radius 7 cm (use $\pi = \frac{22}{7}$). Column B: the perimeter of a square with side 11 cm. Compare the two columns.

$A = 2 \times \frac{22}{7} \times 7 = 44$ cm. $B = 4 \times 11 = 44$ cm. $A = B$.

47 Column A: the area of a triangle with base 10 and height 8. Column B: the area of a rectangle with length 9 and width 5. Compare the two columns.

$A = \frac{1}{2} \times 10 \times 8 = 40$. $B = 9 \times 5 = 45$. $B > A$.

48 Column A: the sum of the interior angles of a quadrilateral. Column B: three times the sum of the interior angles of a triangle. Compare the two columns.

$A = 360^\circ$. $B = 3 \times 180^\circ = 540^\circ$. $B > A$.